

Digital signal processing

October 8, 2026

Contents

1	Conventions	2
2	DFT	2
3	FFT	6
4	Finite impulse response filters	7
4.1	Convolution theorem	7
4.2	Ideal LPF	9
5	Infinite impulse response filters	11
5.1	Stability	12
5.2	Notch filter	21
6	Minimum phase	23
6.1	Group delay	23
6.2	All-pass	26
6.3	All-pass decomposition	28

1 Conventions

Let $f : \mathbb{R} \rightarrow \mathbb{C}$ be a 2π -periodic, integrable function. The Fourier coefficients of f are defined by

$$\hat{f}(k) = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-ikx} dx, \quad k \in \mathbb{Z}.$$

The Fourier series of f is given by

$$f(x) = \sum_{k \in \mathbb{Z}} \hat{f}(k) e^{ikx}$$

For 2π -periodic functions $f, g : \mathbb{R} \rightarrow \mathbb{C}$ we define the circular convolution by

$$(f * g)(x) = \int_0^{2\pi} f(y) g(x - y) dy \quad (1)$$

The root of unity:

$$\omega_N = e^{\frac{2\pi i}{N}}$$

2 DFT

Lemma 2.1. The following holds for roots of unity:

2.1.1 $\omega_N^{kN} = 1 \quad \forall k \in \mathbb{Z}$

2.1.2 N even. $\omega_{N/2} = \omega_N^2$

2.1.3 $\overline{\omega_N} = \omega_N^{-1}$

2.1.4 $\omega_N^{kN/2} = (-1)^k \quad \forall k \in \mathbb{Z}$

Proof.

1.

$$\omega_N^N = \left(e^{\frac{2\pi i}{N}} \right)^N = e^{2\pi i} = \cos(2\pi) + i \sin(2\pi) = 1$$

$$\omega_N^{kN} = \left(\omega_N^N \right)^k = 1^k = 1$$

2. $\omega_{N/2} = e^{\frac{2\pi i}{N/2}} = e^{2 \cdot \frac{2\pi i}{N}} = \left(e^{\frac{2\pi i}{N}} \right)^2 = \omega_N^2$

3. $\overline{\omega_N} = e^{\overline{\frac{2\pi i}{N}}} = e^{-\frac{2\pi i}{N}} = \left(e^{\frac{2\pi i}{N}} \right)^{-1} = \omega_N^{-1}$

4. $\omega_N^{k \cdot N/2} = \left(e^{2\pi i \frac{N/2}{N}} \right)^k = (e^{\pi i})^k = (-1)^k$

□

Suppose we have $f : \mathbb{R} \rightarrow \mathbb{C}$ which is 2π -periodic. Consider the values of this function at points $\frac{0 \cdot 2\pi}{N}, \frac{1 \cdot 2\pi}{N}, \dots, \frac{(N-1) \cdot 2\pi}{N}$.

$$x_j := \frac{j \cdot 2\pi}{N}$$

$$y_j := f(x_j) = \sum_{k=-\infty}^{\infty} \hat{f}(k) e^{ikx_j} = \sum_{k=-\infty}^{\infty} \hat{f}(k) e^{2\pi i \frac{kj}{N}} = \sum_{k=-\infty}^{\infty} \hat{f}(k) \omega_N^{kj}$$

By Lemma 2.1.1 we have that $\omega_N^{(k+nN)j} = \omega_N^{kj} \omega_N^{(nj)N} = \omega_N^{kj}$

By grouping the terms with $k_1 = k_2 \pmod{N}$ and obtain the following:

$$y_j = \sum_{k=0}^{N-1} \omega_N^{kj} \left(\sum_{n=-\infty}^{\infty} \hat{f}(k + nN) \right) = \sum_{k=0}^{N-1} \omega_N^{kj} z_k \quad (2)$$

We define the matrix $F_{kj} = \frac{1}{\sqrt{N}} \omega_N^{kj}$. Then:

$$y_j = \sqrt{N} (Fz)_j \Rightarrow y = \sqrt{N} Fz \Rightarrow z = \frac{1}{\sqrt{N}} F^{-1} y = Ty \quad (3)$$

Later we use the vector z to indicate the output of the DFT, namely $F^{-1}y$

Theorem 2.1. *The matrix $F_{kj} = \frac{1}{\sqrt{N}} \omega_N^{kj}$ is unitary*

Proof.

$$\begin{aligned} (F \overline{F^T})_{jl} &= \frac{1}{\sqrt{N}} \overline{\left(\frac{1}{\sqrt{N}} \right)} \sum_{k=0}^{N-1} F_{jk} (\overline{F^T})_{kl} = \frac{1}{N} \sum_{k=0}^{N-1} F_{jk} \overline{F_{lk}} = \\ &= \frac{1}{N} \sum_{k=0}^{N-1} \omega_N^{jk} \overline{\omega_N^{lk}} \stackrel{(a)}{=} \frac{1}{N} \sum_{k=0}^{N-1} \omega_N^{jk} \omega_N^{-lk} = \frac{1}{N} \sum_{k=0}^{N-1} \omega_N^{(j-l)k} \end{aligned}$$

In (a) we used the identity $\overline{\omega_N^{lk}} = \omega_N^{-lk}$ from Lemma 2.1.3

1. $j \neq l$. Using the formula for sum of geometric series:

$$\sum_{k=0}^{N-1} \omega_N^{(j-l)k} = \frac{1 - \left(\omega_N^{j-l} \right)^N}{1 - \omega_N^{j-l}} = \frac{1 - \left(\omega_N^N \right)^{j-l}}{1 - \omega_N^{j-l}} \stackrel{(a)}{=} \frac{1 - 1^{j-l}}{1 - \omega_N^{j-l}} = 0$$

In (a) we used the identity $\omega_N^N = 1$ from Lemma 2.1.1. The condition $j \neq l$ ensures that the denominator is non-zero, so the expression is well-defined.

2. $j = l$. Then:

$$\sum_{k=0}^{N-1} \omega_N^{(j-l)k} = \sum_{k=0}^{N-1} 1 = N \Rightarrow (F \overline{F^T})_{jl} = \frac{1}{N} N = 1$$

The matrix $FF^* = I$, so F unitary and $F^{-1} = \overline{F^T}$

The entries of F^{-1} are given by $F_{ij}^{-1} = (\overline{F_{ji}}) = \frac{1}{\sqrt{N}} \overline{\omega_N^{ji}} = \frac{1}{\sqrt{N}} \omega_N^{-ij}$. □

Using the result of Theorem 2.1 and equations (3), (2) we define the discrete Fourier transform:

$$z_j = (F^{-1}y)_j = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-kj} y_k \quad (\text{DFT})$$

$$y_j = (Fz)_j = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{kj} z_k \quad (\text{IDFT})$$

Remark 2.1. Now we make two remarks about the DFT structure:

2.1.1 Firstly using DFT of size n we are unable to detect the frequencies above the $(n-1)$ -th one. In fact as the equation (2) suggests, the frequency of the form $k + nN$ with $k \in 0, \dots, N-1$ contributes to the entry with index k . However the maximum frequency human ear can process is approximately 20000, so this issue is handled by choosing an appropriate sample rate, that is usually 44100hz, which is double the maximum frequency which can be detected. The exact reason for it such choice is discussed later. The following plot illustrates the issue discussed (so-called aliasing):

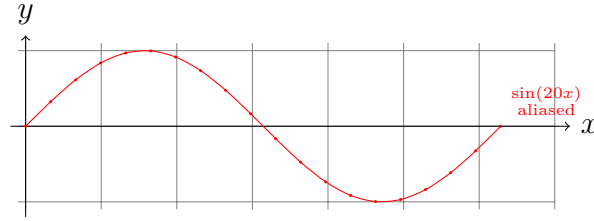


Figure 1: 20 discrete points

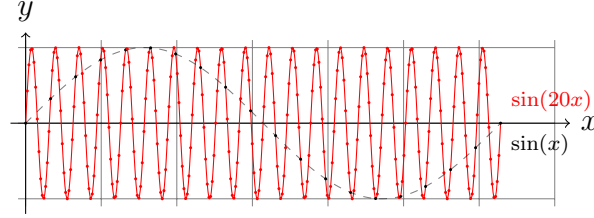


Figure 2: 286 discrete points

2.1.2 The meaning of the phase is often overlooked in the literature, so we consider an example: Let $y = e_k + e_{N-k}$. Then:

$$\begin{aligned} Ty = T_{*k} + T_{*(N-k)} &= \begin{pmatrix} \omega_N^0 \\ \omega_N^{-k} \\ \vdots \\ \omega_N^{-k(N-1)} \end{pmatrix} + \begin{pmatrix} \omega_N^0 \\ \omega_N^{-(N-k)} \\ \vdots \\ \omega_N^{-(N-k)(N-1)} \end{pmatrix} \\ &\stackrel{\text{Lemma 2.1.1}}{=} \begin{pmatrix} \omega_N^0 + \omega_N^0 \\ \omega_N^{-k} + \omega_N^k \\ \vdots \\ \omega_N^{-k(N-1)} + \omega_N^{k(N-1)} \end{pmatrix} = \begin{pmatrix} 2 \cos(0) \\ 2 \cos\left(\frac{2\pi k}{N}\right) \\ \vdots \\ 2 \cos\left(\frac{2\pi k(N-1)}{N}\right) \end{pmatrix} \end{aligned}$$

As we see from the computation, the output is the function $\cos(kx)$ discretized at points $x_j = \frac{2\pi j}{N}$ for $j \in \{0, \dots, N-1\}$. What happens if we multiply the positive frequency component of y by ω_N^{-k} and the negative one by ω_N^k ?

$$\begin{aligned} y^* &= \omega_N^{-k} e_k + (\omega_N^k e_{N-k}) \\ Ty^* &= \begin{pmatrix} \omega_N^{-k} \omega_N^0 + \omega_N^k \omega_N^0 \\ \omega_N^{-k} \omega_N^{-k} + \omega_N^k \omega_N^k \\ \vdots \\ \omega_N^{-k} \omega_N^{-k(N-1)} + \omega_N^k \omega_N^{k(N-1)} \end{pmatrix} \\ &= \begin{pmatrix} \omega_N^{-k} + \omega_N^k \\ \omega_N^{-2k} + \omega_N^{2k} \\ \vdots \\ \omega_N^{-kN} + \omega_N^{kN} \end{pmatrix} = \begin{pmatrix} 2 \cos\left(\frac{2\pi k}{N}\right) \\ 2 \cos\left(\frac{2\pi \cdot 2k}{N}\right) \\ 2 \cos\left(\frac{2\pi \cdot 3k}{N}\right) \\ \vdots \\ 2 \cos\left(\frac{2\pi kN}{N}\right) \end{pmatrix} \end{aligned}$$

We effectively moved the phase of the cosine by $\frac{2\pi}{N}$, since

$$\begin{aligned} (Ty^*)_j &= 2 \cos\left(\frac{2\pi k(j+1)}{N}\right) = 2 \cos\left(k\left(\frac{2\pi j}{N} + \frac{2\pi}{N}\right)\right) \\ &= 2 \cos\left(k\left(x_j + \frac{2\pi}{N}\right)\right) \end{aligned}$$

not hard yet difficult to see by yourself (at least for me)

Theorem 2.2 (Real signal conjugate symmetry). *Let N be even. The following are equivalent:*

1. $y = \bar{y}, y \in \mathbb{C}^N$
2. $z_j = \overline{z_{N-j}} \quad \forall j \in \{1, \dots, N/2 - 1\}$ and $z_0, z_{N/2} \in \mathbb{R}$

Proof.

”(1) $\Rightarrow$ (2)” Using the formula for [DFT](#)

$$\begin{aligned} z_j &= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-kj} y_k = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \overline{\omega_N^{kj} y_k} \stackrel{y=\bar{y}}{=} \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \overline{\omega_N^{kj} y_k} = \\ &= \overline{\left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{kj} y_k \right)} \stackrel{\omega_N^{kN}=1}{=} \overline{\left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-kN} \omega_N^{kj} y_k \right)} = \\ &= \overline{\left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-k(N-j)} y_k \right)} = \overline{z_{N-j}} \\ z_0 &= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-k \cdot 0} y_k = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} y_k \in \mathbb{R} \end{aligned}$$

By Lemma 2.1.4:

$$z_{N/2} = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-kN/2} y_k = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} (-1)^k y_k \in \mathbb{R}$$

”(2) $\Rightarrow$ (1)” Using the formula for IDFT

$$\begin{aligned} y_j &= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{kj} z_k = \frac{1}{\sqrt{N}} \left(z_0 + (-1)^j z_{N/2} + \sum_{k=1}^{N/2-1} \omega_N^{kj} z_k + \sum_{l=N/2+1}^{N-1} \omega_N^{lj} z_l \right) \\ &\quad \sum_{k=1}^{N/2-1} \omega_N^{kj} z_k + \sum_{l=N/2+1}^{N-1} \omega_N^{lj} z_l \stackrel{l'=N-l}{=} \sum_{k=1}^{N/2-1} \omega_N^{kj} z_k + \sum_{l=1}^{N/2-1} \omega_N^{(N-l)j} z_{N-l} = \\ &\quad \sum_{k=1}^{N/2-1} \left(\omega_N^{kj} z_k + \omega_N^{(N-k)j} z_{N-k} \right) = \sum_{k=1}^{N/2-1} \left(\omega_N^{kj} z_k + \overline{\omega_N^{kj} z_k} \right) \in \mathbb{R} \end{aligned} \quad (4)$$

By plugging this expression back to (4) we get:

$$(4) = \frac{1}{\sqrt{N}} \left(z_0 + (-1)^j z_{N/2} + \sum_{k=1}^{N/2-1} \left(\omega_N^{kj} z_k + \overline{\omega_N^{kj} z_k} \right) \right) \in \mathbb{R}$$

since all the summands and the factor in front are real. $\square$

3 FFT

Theorem 3.1. Let $y \in \mathbb{C}^N$ and N is even. $K = N/2$. Let:

$$\begin{aligned} y^{(1)} &= (y_0, y_2, \dots, y_{2K-2})^T & y^{(2)} &= (y_1, y_3, \dots, y_{2K-1})^T \\ z^{(1)} &= F_K^{-1} y^{(1)} & z^{(2)} &= F_K^{-1} y^{(2)} \end{aligned}$$

Then:

$$\begin{aligned} z_j &= \frac{1}{\sqrt{2}} \left(z_j^{(1)} + \omega_N^{-j} z_j^{(2)} \right) \quad \forall j \in \{0, \dots, K-1\} \\ z_{j+K} &= \frac{1}{\sqrt{2}} \left(z_j^{(1)} - \omega_N^{-j} z_j^{(2)} \right) \quad \forall j \in \{0, \dots, K-1\} \end{aligned}$$

where $z = F_N^{-1} y$

Proof.

$$\begin{aligned} z_j &= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-kj} y_k = \frac{1}{\sqrt{N}} \left(\sum_{\substack{k=0 \\ k \text{ even}}}^{N-1} \omega_N^{-kj} y_k + \sum_{\substack{k=0 \\ k \text{ odd}}}^{N-1} \omega_N^{-kj} y_k \right) = \\ &\quad \frac{1}{\sqrt{N}} \left(\sum_{k=0}^{K-1} \omega_N^{-2kj} y_{2k} + \sum_{k=0}^{K-1} \omega_N^{-(2k+1)j} y_{2k+1} \right) \stackrel{\text{Lemma 2.1.2}}{=} \\ &\quad \frac{1}{\sqrt{N}} \left(\sum_{k=0}^{K-1} \omega_K^{-kj} y_{2k} + \omega_N^{-j} \sum_{k=0}^{K-1} \omega_K^{-kj} y_{2k+1} \right) = \\ &\quad \frac{1}{\sqrt{2}\sqrt{K}} \left(\sum_{k=0}^{K-1} \omega_K^{-kj} y_{2k} + \omega_N^{-j} \sum_{k=0}^{K-1} \omega_K^{-kj} y_{2k+1} \right) \end{aligned} \quad (5)$$

For $j \in \{0, \dots, K-1\}$ we get the claim immediately. We now focus on the case, when $j = j' + K \in \{K, \dots, N-1\}$. The equation (5) becomes:

$$\begin{aligned} \sqrt{2K}z_j &= \sum_{k=0}^{K-1} \omega_K^{-k(j'+K)} y_{2k} + \omega_N^{-(j'+K)} \sum_{k=0}^{K-1} \omega_K^{-k(j'+K)} y_{2k+1} = \\ &= \sum_{k=0}^{K-1} \omega_K^{-kj'} y_{2k} + \omega_N^{-j'} \omega_N^{-K} \sum_{k=0}^{K-1} \omega_K^{-kj'} y_{2k+1} \stackrel{\text{Lemma 2.1.4}}{=} \\ &= \sum_{k=0}^{K-1} \omega_K^{-kj'} y_{2k} - \omega_N^{-j'} \sum_{k=0}^{K-1} \omega_K^{-kj'} y_{2k+1} \end{aligned}$$

which proves the second part of the claim. $\square$

Algorithm 1: FFT

Data: $y \in \mathbb{C}^N$, N is a power of two, $n > 1$

if $N > n$ **then**

$z^{(1)} \leftarrow \text{FFT}(y^{(1)});$

$z^{(2)} \leftarrow \text{FFT}(y^{(2)});$

 /* $N/2$ is a power of two

*/

$z_j = \frac{1}{\sqrt{2}} \left(z_j^{(1)} + \omega_N^{-j} z_j^{(2)} \right) \quad \forall j \in \{0, \dots, K-1\};$

$z_{j+K} = \frac{1}{\sqrt{2}} \left(z_j^{(1)} - \omega_N^{-j} z_j^{(2)} \right) \quad \forall j \in \{0, \dots, K-1\};$

return z ;

else

return $F^{-1}y$;

 /* see the expression DFT

*/

4 Finite impulse response filters

4.1 Convolution theorem

Lemma 4.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ have countably many discontinuities. Then f is measurable

Proof. 1. Let $D \subset \mathbb{R}$ be the set of all discontinuities of f . We have that D has countably many elements. For any set $M \in \mathcal{B}$ we have the following:

$$\begin{aligned} f^{-1}(M) &= (f^{-1}(M) \cap D) \cup (f^{-1}(M) \setminus D) \\ &= (f^{-1}(M) \cap D) \cup \left(\left(f|_{\mathbb{R} \setminus D} \right)^{-1}(M) \right) \end{aligned} \tag{6}$$

We have $(f^{-1}(M) \cap D) \subset D$, so it is countable and hence is Borel measurable, $f|_{\mathbb{R} \setminus D}$ is a measurable mapping since $\mathbb{R} \setminus D$ is a Borel set and the restriction is continuous on its domain. Thus, $f^{-1}(M)$ is the union of two measurable sets, so is measurable as well. That proves measurability of f . $\square$

Remark 4.1. The statement of the Lemma 4.1 can be relaxed to the set of discontinuities having measure zero. Let F be a generalized distribution function (monotonous and right continuous). Then the following mapping defines an outer measure:

$$\mu^* : \mathcal{P}(\mathbb{R}) \rightarrow [0; \infty], A \mapsto \inf \left\{ \sum_{n=1}^{\infty} F(b_n) - F(a_n) : A \subset \bigcup_{n=1}^{\infty} (a_n; b_n], a_n < b_n \forall n \in \mathbb{N} \right\}$$

The Lebesgue measure is $\mu = \mu^*|_{\mathcal{B}}$ for $F(x) = x$. However we know that $\mathcal{B} \subset \mathcal{L} := \{A \subset \mathbb{R} : A \text{ is } \mu^* \text{ measurable}\}$. However we cannot guarantee $\mathcal{B} = \mathcal{L}$ ($\mathcal{L}$ is a sigma algebra by Caratheodori restriction theorem)

Consider a set $N \subset \mathbb{R}$ with $\mu^*(N) = 0$. For any $B \subset \mathbb{R}$:

$$\mu^*(B) = \mu^*((B \cap N) \cup (B \cap N^C)) \leq \mu^*(B \cap N) + \mu^*(B \cap N^C)$$

by sigma subadditivity of μ^* . Moreover by monotonicity of μ^* we have that $\mu^*(B \cap N) \leq \mu^*(N) = 0$, so $\mu^*(B \cap N^C) \geq \mu^*(B)$. By applying monotonicity again $\mu^*(B \cap N^C) \leq \mu^*(B)$, so $\mu^*(B \cap N^C) = \mu^*(B)$. By combining those results together we obtain $\mu^*(B) = \mu^*(B \cap N) + \mu^*(B \cap N^C)$, so $N \in \mathcal{M}$

Now take a look at (6):

$$f^{-1}(M) = (f^{-1}(M) \cap D) \cup \left((f|_{\mathbb{R} \setminus D})^{-1}(M) \right)$$

The set $f^{-1}(M) \cap D \subset D$, so it is a null set and hence is in $\mathcal{L}$. Also we know that $(f|_{\mathbb{R} \setminus D})^{-1}(M) \subset \mathcal{B} \subset \mathcal{L}$, which implies $f^{-1}(M) \subset \mathcal{L}$. Therefore f is $\mathcal{L} \rightarrow \mathcal{B}$ measurable.

Theorem 4.1 (Convolution theorem). *Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be 2π periodic piecewise continuous and bounded. Then:*

$$\frac{1}{2\pi} \widehat{f * g}(k) = \hat{f}(k) \hat{g}(k)$$

Proof.

$$\begin{aligned} 2\pi \cdot \widehat{f * g}(k) &= \int_0^{2\pi} f * g(x) e^{-ikx} dx \\ &\stackrel{(1)}{=} \int_0^{2\pi} \left(\int_0^{2\pi} f(y) g(x-y) dy \right) \cdot e^{-ikx} dx \\ &= \int_0^{2\pi} \int_0^{2\pi} f(y) g(x-y) e^{-ikx} dy dx \end{aligned} \tag{7}$$

Let $F(x, y) = f(y)g(x-y)e^{-ikx}$. We have that both f, g are measurable by Lemma 4.1 since they have finitely many discontinuities. Therefore $\operatorname{Re}(F) = f(\pi_y(x, y))g(x-y) \cos(kx)$ and $\operatorname{Im}(F) = -f(\pi_y(x, y))g(x-y) \sin(kx)$ are $\mathcal{B}_2 \rightarrow \mathcal{B}$ measurable.

$$|\operatorname{Re}(F(x, y))|, |\operatorname{Im}(F(x, y))| \leq |F(x, y)| = |f(y)g(x-y)| \leq C < \infty$$

since both f and g are bounded.

$$\begin{aligned} \int_{[0, 2\pi]^2} |\operatorname{Re}(F)(x, y)| d\lambda^2(x, y) &\leq \int_{\mathbb{R}^2} C \cdot \mathbf{1}_{[0, 2\pi]^2}(x, y) d\lambda^2(x, y) \\ &= C\lambda^2([0, 2\pi]^2) = 4\pi^2 C < \infty \end{aligned}$$

The same holds for $\text{Im}(F)$. Since the function F is absolutely integrable on $[0, 2\pi]^2$ we can use Fubini's theorem. This theorem also ensures, that $\int_0^{2\pi} f(y)g(x-y)e^{-ikx} dy$ is measurable wherever the integral exist. Moreover, we can say that the integral exists for all y , since the integrand is uniformly bounded.

$$\begin{aligned} \int_0^{2\pi} \int_0^{2\pi} F(x, y) dy dx &= \int_0^{2\pi} \int_0^{2\pi} F(x, y) dx dy \\ &= \int_0^{2\pi} \int_0^{2\pi} f(y)g(x-y)e^{-iky}e^{-ik(x-y)} dx dy \\ &= \int_0^{2\pi} f(y)e^{-iky} \int_0^{2\pi} g(x-y)e^{-ik(x-y)} dx dy \end{aligned}$$

Do the substitution $t = x - y$, $\frac{\partial t}{\partial x} = 1$ (this substitution is a diffeomorphism as a mapping $\Phi : [0; 2\pi] \rightarrow [-y, 2\pi - y]$, $x \mapsto x - y$ with $|\det(J \Phi(x))| = 1$)

$$\begin{aligned} \int_0^{2\pi} f(y)e^{-iky} \int_0^{2\pi} g(x-y)e^{-ik(x-y)} dx dy &= \\ \int_0^{2\pi} f(y)e^{-iky} \int_{-y}^{2\pi-y} g(t)e^{-ikt} dt dy &= \\ \int_0^{2\pi} f(y)e^{-iky} \int_0^{2\pi} g(t)e^{-ikt} dt dy &= \\ 2\pi \int_0^{2\pi} f(y)e^{-iky} \hat{g}(k) dy &= 4\pi^2 \hat{g}(k) \hat{f}(k) \end{aligned}$$

Thus:

$$2\pi \widehat{f * g}(k) = 4\pi^2 \hat{g}(k) \hat{f}(k) \Rightarrow \frac{1}{2\pi} \widehat{f * g}(k) = \hat{g}(k) \hat{f}(k)$$

□

Theorem 4.2 (Discrete convolution theorem). *Let $x, y \in \mathbb{R}^n$. Then the following holds:*

$$\frac{1}{\sqrt{N}}(F(x \circledast y)) = Fx \odot Fy$$

where $\circledast$ denotes the circular convolution, and $\odot$ is the component-wise multiplication

Proof. This is the direct analogy to the continuous case in Theorem 4.1

□

4.2 Ideal LPF

Using this knowledge we can derive an ideal low pass filter.

$$z_i = \begin{cases} 0, & i \in (k; n - k) \\ 1, & \text{otherwise} \end{cases}$$

This is an impulse reponse of the filter cutting all the frequencies above the k -th one. Recall that the $N - k$ element in DFT vector corresponds to the negative oscillation with the frequency k , due to cyclicity of $e^{\frac{2\pi i j x}{N}}$.

Using the formula for [IDFT](#):

$$y_j = \sum_{l=0}^{N-1} F_{jl} z_k = \sum_{l=0}^k F_{jl} + \sum_{l=N-k}^{N-1} F_{jl} = \frac{1}{\sqrt{N}} \left(\sum_{l=0}^k \omega_N^{jl} + \sum_{l=N-k}^{N-1} \omega_N^{jl} \right)$$

Substitute $l = N - m - 1$ in the second sum:

$$\begin{aligned} \frac{1}{\sqrt{N}} \left(\sum_{l=0}^k \omega_N^{jl} + \sum_{l=N-k}^{N-1} \omega_N^{jl} \right) &= \frac{1}{\sqrt{N}} \left(\sum_{l=0}^k \omega_N^{jl} + \sum_{m=0}^{k-1} \omega_N^{j(N-m-1)} \right) = \\ \frac{1}{\sqrt{N}} \left(\sum_{l=0}^k \omega_N^{jl} + \sum_{m=0}^{k-1} \omega_N^{j(N-m-1)} \right) &= \frac{1}{\sqrt{N}} \left(\sum_{l=0}^k \omega_N^{jl} + \sum_{m=0}^{k-1} \omega_N^{jN} \omega_N^{-j} \omega_N^{-jm} \right) \stackrel{2.1.1}{=} \\ &= \frac{1}{\sqrt{N}} \left(\omega_N^{jk} + \sum_{l=0}^{k-1} \omega_N^{jl} + \omega_N^{-j} \sum_{m=0}^{k-1} \omega_N^{-jm} \right) \end{aligned}$$

Let $j \neq 0$. Using the sum of geometric series:

$$\sum_{l=0}^{k-1} \omega_N^{jl} = \sum_{l=0}^{k-1} (\omega_N^j)^l = \frac{1 - (\omega_N^j)^k}{1 - \omega_N^j}$$

Similarly:

$$\begin{aligned} \omega_N^{-j} \sum_{m=0}^{k-1} \omega_N^{-jm} &= \omega_N^{-j} \sum_{m=0}^{k-1} (\omega_N^{-j})^m = \frac{\omega_N^{-j} - \omega_N^{-j} (\omega_N^{-j})^k}{1 - \omega_N^{-j}} = \\ \frac{\omega_N^{-j} - \omega_N^{-j} (\omega_N^{-j})^k}{1 - \omega_N^{-j}} \frac{\omega_N^j}{\omega_N^j} &= \frac{1 - (\omega_N^{-j})^k}{\omega_N^j - 1} = \frac{(\omega_N^{-j})^k - 1}{1 - \omega_N^j} \end{aligned}$$

By combining the terms together:

$$\sum_{l=0}^{k-1} \omega_N^{jl} + \omega_N^{-j} \sum_{l=0}^{k-1} \omega_N^{-jl} = \frac{1 - (\omega_N^j)^k}{1 - \omega_N^j} + \frac{(\omega_N^{-j})^k - 1}{1 - \omega_N^j} = \frac{\omega_N^{-jk} - \omega_N^{jk}}{1 - \omega_N^j}$$

This yields the following expression for y_j , when $j \neq 0$:

$$y_j = \frac{1}{\sqrt{N}} \left(\omega_N^{jk} + \frac{\omega_N^{-jk} - \omega_N^{jk}}{1 - \omega_N^j} \right)$$

If $j = 0$, then all the summands are equal to one, so get just $(2k + 1)/\sqrt{N}$. By the Theorem [4.2](#):

$$\frac{1}{\sqrt{N}} (F(x \circledast y)) = Fx \odot Fy$$

which allows us to implement the low pass filter using its impulse response. That is the point, where theory meets the real world example and makes use of the abstract theorem. In general this concept can be generalized to any signal, which can be used, for instance for reverb implementation if we have an impulse response of a room (a recording of echo induced by a short impact).

Now we take a closer look on the expression we have derived above:

$$e^{-ix} - e^{ix} = 2 \operatorname{Im}(e^{-ix}) = 2i \sin(-x) = -2i \sin(x)$$

$$1 - e^{ix} = e^{\frac{ix}{2}} \left(e^{-\frac{ix}{2}} - e^{\frac{ix}{2}} \right) = -2i \cdot e^{\frac{ix}{2}} \sin\left(\frac{x}{2}\right)$$

Therefore:

$$\begin{aligned} 1 - \omega_N^j &= 1 - e^{2\pi i \frac{j}{N}} = -2i \cdot e^{\frac{\pi i \cdot j}{N}} \sin\left(\frac{\pi j}{N}\right) \\ \omega_N^{-jk} - \omega_N^{jk} &= e^{-2\pi i \frac{kj}{N}} - e^{2\pi i \frac{kj}{N}} = -2i \sin\left(\frac{2\pi kj}{N}\right) \\ \frac{\omega_N^{-jk} - \omega_N^{jk}}{1 - \omega_N^j} &= \frac{-2i \sin\left(\frac{2\pi kj}{N}\right)}{-2i \cdot e^{\frac{\pi i \cdot j}{N}} \sin\left(\frac{\pi j}{N}\right)} = \frac{e^{-\frac{\pi i \cdot j}{N}} \sin\left(\frac{2\pi kj}{N}\right)}{\sin\left(\frac{\pi j}{N}\right)} \end{aligned}$$

With $x_j = 2\pi j/N$ we have:

$$\frac{e^{-\frac{\pi i \cdot j}{N}} \sin\left(\frac{2\pi kj}{N}\right)}{\sin\left(\frac{\pi j}{N}\right)} = \frac{e^{-ix_j/2} \sin(kx_j)}{\sin(x_j/2)} = \frac{\cos(x_j/2) \sin(kx_j)}{\sin(x_j/2)} - i \sin(kx_j)$$

Add the term ω_N^{jk} we had previously:

$$\begin{aligned} \omega_N^{jk} + \frac{\cos(x_j/2) \sin(kx_j)}{\sin(x_j/2)} - i \sin(kx_j) &= \\ \cos(kx_j) + i \sin(kx_j) + \frac{\cos(x_j/2) \sin(kx_j)}{\sin(x_j/2)} - i \sin(kx_j) &= \\ \cos(kx_j) + \frac{\cos(x_j/2) \sin(kx_j)}{\sin(x_j/2)} &= \\ \frac{\cos(kx_j) \sin(x_j/2) + \cos(x_j/2) \sin(kx_j)}{\sin(x_j/2)} &= \end{aligned} \tag{8}$$

Using the equality $\sin(x+y) = \sin(x) \cos(y) + \sin(y) \cos(x)$:

$$\begin{aligned} (8) &= \frac{\sin((k+1/2)x_j)}{\sin(x_j/2)} \\ y_j &= \begin{cases} \frac{1}{\sqrt{N}} \frac{\sin((k+1/2)x_j)}{\sin(x_j/2)}, & j \in \{1, \dots, N-1\} \\ (2k+1)/\sqrt{N}, & j = 0 \end{cases} \end{aligned}$$

Miraculously we got a continuous Dirichlet kernel, discretized at points x_j . One can show, that the impulse response of Dirichlet kernel in continuous space acts in the exact same way, namely cuts the frequencies above the k -th one. As we had seen in equation (2), $z_k = \sum_{n \in \mathbb{Z}} \hat{f}(k + nN)$, so given that $\widehat{D}_n(k) = \begin{cases} 1, & k \leq n \\ 0, & \text{otherwise} \end{cases}$, this result was not completely unexpected.

5 Infinite impulse response filters

We kick off with the example

Example 5.1. Consider the sequence $x[n] \in \mathbb{R}$, which represents the signal evolving in time. Let:

$$y[0] = x[0], \quad y[n] = x[n] + \alpha y[n-1] \quad \forall n > 0$$

The structure of the formula suggests, that we should use an induction to get a closed formula for y :

Base. $y[0] = x[0] = \alpha^0 x[0]$

Step. Assume that it holds for some n , that

$$y[n] = \sum_{j=0}^n \alpha^{n-j} x[j] \quad (9)$$

Then:

$$\begin{aligned} y[n+1] &= x[n+1] + \alpha y[n] = \alpha^0 x[n+1] + \alpha \sum_{j=0}^n \alpha^{n-j} x[j] \\ &= \alpha^0 x[n+1] + \sum_{j=0}^n \alpha^{n+1-j} x[j] = \sum_{j=0}^{n+1} \alpha^{n+1-j} x[j] \end{aligned}$$

Note that the formula (9) can be written as a following convolution:

$$y[n] = \sum_{j=0}^n \alpha^{n-j} x[j] = \sum_{j=0}^n h_n[n-j] x[j] = x * h_n$$

Thus:

$$h_n = (\alpha^0, \alpha^1, \dots, \alpha^n)^T$$

As in the case of Ideal low pass filter in Section 4.2 we got an impulse response of the filter introduced above. The only difference is that this filter has an impulse response with the size proportional to the sample size. That said the IR size grows infinitely in time unlike the IR of an ideal LPF, which has the same size for each frame.

5.1 Stability

Let us consider the general form of the filters we would like to analyze:

$$y[n] = \sum_{i=0}^N b_i x[n-i] + \sum_{i=1}^M a_i y[n-i] \quad (10)$$

Theorem 5.1 (Discrete Tonelli). *Let us have the sequence $x_{nk} \in [0; \infty]$. Then:*

$$\sum_{k=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} x_{nk} = \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x_{nk}$$

Proof. We have that the space $(\mathbb{Z}, \mathcal{P}(\mathbb{Z}), \mu)$ is σ -finite with the counting measure μ . This is easily verified using that $\mathbb{Z} = \bigcup_{k \in \mathbb{Z}} \{k\}$. Therefore the product measure $\mu \otimes \mu$ is well-defined and unique.

Define

$$f : \mathbb{Z} \times \mathbb{Z} \rightarrow [0; \infty], \quad (n, k) \mapsto x_{nk}$$

We define the function slices as:

$$\begin{aligned} f_n : \mathbb{Z} &\rightarrow [0; \infty], & k &\mapsto x_{nk} \\ f^{(k)} : \mathbb{Z} &\rightarrow [0; \infty], & n &\mapsto x_{nk} \end{aligned}$$

And we observe that $\mathbb{1}_{[-K,K]} \cdot f_n = \sum_{k=-K}^K \mathbb{1}_{\{k\}} \cdot x_{nk} \uparrow f_n$, so using the the fact that each $\mathbb{1}_{[-K,K]} \cdot f_n$ is simple:

$$\int_{\mathbb{Z}} \mathbb{1}_{[-K,K]} \cdot f_n \, d\mu(k) = \sum_{k=-K}^K \mu(\{k\}) x_{nk} = \sum_{k=-K}^K x_{nk}$$

Using Monotone convergence theorem:

$$\begin{aligned} \sum_{k=-\infty}^{\infty} x_{nk} &= \lim_{K \rightarrow \infty} \int_{\mathbb{Z}} \mathbb{1}_{[-K,K]} \cdot f_n \, d\mu(k) \\ &= \int_{\mathbb{Z}} \lim_{K \rightarrow \infty} (\mathbb{1}_{[-K,K]} \cdot f_n) \, d\mu(k) = \int_{\mathbb{Z}} f_n(k) \, d\mu(k) \end{aligned}$$

Using the similar construction:

$$g : \mathbb{Z} \rightarrow [0; \infty], \quad n \rightarrow \int_{\mathbb{Z}} f_n(k) \, d\mu(k)$$

And its approximation:

$$\mathbb{1}_{[-N,N]} \cdot g = \sum_{n=-N}^N \left(\mathbb{1}_{\{n\}} \int_{\mathbb{Z}} f_n(k) \, d\mu(k) \right) \uparrow g$$

since the integral of the positive function f_n is positive as well, so $g \geq 0$.

$$\begin{aligned} \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x_{nk} &= \lim_{N \rightarrow \infty} \int_{\mathbb{Z}} \mathbb{1}_{[-N,N]} \cdot g(n) \, d\mu(n) \\ &= \int_{\mathbb{Z}} \lim_{N \rightarrow \infty} (\mathbb{1}_{[-N,N]} \cdot g(n)) \, d\mu(n) \\ &= \int_{\mathbb{Z}} g(n) \, d\mu(n) = \int_{\mathbb{Z}} \int_{\mathbb{Z}} f_n(k) \, d\mu(k) \, d\mu(n) \end{aligned}$$

Using Tonelli's theorem we have:

$$\int_{\mathbb{Z} \times \mathbb{Z}} f \, d\mu \otimes \mu(n, k) = \int_{\mathbb{Z}} \int_{\mathbb{Z}} f_n(k) \, d\mu(k) \, d\mu(n) = \int_{\mathbb{Z}} \int_{\mathbb{Z}} f^{(k)}(n) \, d\mu(n) \, d\mu(k)$$

In a similar fashion we can show for the slice at a height k :

$$\sum_{k=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} x_{nk} = \int_{\mathbb{Z}} \int_{\mathbb{Z}} f^{(k)}(n) \, d\mu(n) \, d\mu(k)$$

Which proves:

$$\sum_{k=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} x_{nk} = \int_{\mathbb{Z} \times \mathbb{Z}} f \, d\mu \otimes \mu(n, k) = \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x_{nk}$$

□

Theorem 5.2 (Discrete Fubini). *Let us have the sequence $z_{nk} \in \mathbb{C}$ with*

$$\sum_{(n,k) \in \mathbb{Z} \times \mathbb{Z}} |z_{nk}| < \infty$$

Then:

$$\sum_{k=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} z_{nk} = \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} z_{nk}$$

Proof. Let $x_{nk} \in \mathbb{R}$ and define:

$$x_{nk}^+ = \begin{cases} x_{nk}, & x_{nk} \geq 0 \\ 0, & x_{nk} < 0 \end{cases} \quad x_{nk}^- = \begin{cases} -x_{nk}, & x_{nk} < 0 \\ 0, & x_{nk} \geq 0 \end{cases}$$

Recall that if both sums $\sum_{n=-\infty}^{\infty} a_n$ and $\sum_{n=-\infty}^{\infty} b_n$ with $a_n, b_n \in \mathbb{R}$ exist then:

$$\begin{aligned} \sum_{n=-\infty}^{\infty} (a_n - b_n) &= \lim_{N \rightarrow \infty} \left(\sum_{n=-N}^N (a_n - b_n) \right) \\ &= \lim_{N \rightarrow \infty} \left(\sum_{n=-N}^N a_n - \sum_{n=-N}^N b_n \right) \\ &= \lim_{N \rightarrow \infty} \left(\sum_{n=-N}^N a_n \right) - \lim_{N \rightarrow \infty} \left(\sum_{n=-N}^N b_n \right) \\ &= \sum_{n=-\infty}^{\infty} a_n - \sum_{n=-\infty}^{\infty} b_n \end{aligned}$$

We have that:

$$\left| \sum_{k=-\infty}^{\infty} x_{nk}^{\pm} \right| = \sum_{k=-\infty}^{\infty} x_{nk}^{\pm} \leq \sum_{k=-\infty}^{\infty} |x_{nk}| \quad \forall k \in \mathbb{Z} < \infty$$

so using the comparison principle we have the following:

$$\sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x_{nk}^{\pm} < \infty$$

Therefore:

$$\sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x_{nk} = \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} (x_{nk}^+ - x_{nk}^-) = \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x_{nk}^+ - \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x_{nk}^-$$

The sum order can be interchanged by Theorem 5.1:

$$\begin{aligned} \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x_{nk}^+ - \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x_{nk}^- &= \sum_{k=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} x_{nk}^+ - \sum_{k=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} x_{nk}^- \\ &= \sum_{k=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} (x_{nk}^+ - x_{nk}^-) \\ &= \sum_{k=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} x_{nk} \end{aligned}$$

This argument can be generalized for $z_{nk} \in C$ using the auxiliary sequences $r_{nk} = \text{Re}(z_{nk})$ and $s_{nk} = \text{Im}(z_{nk})$. Both are absolutely summable since $|r_{nk}|, |s_{nk}| < |z_{nk}|$. This immediately results in the desired result applying the previous argumentation to r_{nk} and s_{nk} . $\square$

Lemma 5.1 (Z-transform properties). Let $x \xleftrightarrow{\mathcal{Z}} X(z)$ and $y \xleftrightarrow{\mathcal{Z}} Y(z)$. The following holds:

$$5.1.1 \quad x[n-k] \xleftrightarrow{\mathcal{Z}} z^{-k}X(z)$$

$$5.1.2 \quad \alpha x[n] + \beta y[n] \xleftrightarrow{\mathcal{Z}} \alpha X(z) + \beta Y(z)$$

$$5.1.3 \quad x[-n] \xleftrightarrow{\mathcal{Z}} \alpha X(z^{-1})$$

Proof.

1.

$$\begin{aligned} \sum_{n=-\infty}^{\infty} x[n-k]z^{-n} &= \sum_{m=-\infty}^{\infty} x[m]z^{-(m+k)} \\ &= z^{-k} \sum_{m=-\infty}^{\infty} x[m]z^{-m} \\ &= z^{-k}X(z) \end{aligned}$$

2. We assume that both $X(z)$ and $Y(z)$ converge for the given value of z

$$\begin{aligned} \sum_{n=-\infty}^{\infty} (\alpha x[n] + \beta y[n])z^{-n} &= \sum_{n=-\infty}^{\infty} \alpha x[n]z^{-n} + \beta y[n]z^{-n} \\ &= \alpha \sum_{n=-\infty}^{\infty} x[n]z^{-n} + \beta \sum_{n=-\infty}^{\infty} y[n]z^{-n} \\ &= \alpha X(z) + \beta Y(z) \end{aligned}$$

3.

$$\sum_{n=-\infty}^{\infty} x[-n]z^{-n} \underset{k=-n}{=} \sum_{k=-\infty}^{\infty} x[k]z^k = \sum_{k=-\infty}^{\infty} x[n](z^{-1})^{-k} = X(z^{-1})$$

$\square$

Taking the Z-transform of (10):

$$y[n] \xleftrightarrow{\mathcal{Z}} \sum_{i=0}^N b_i z^{-i} X(z) + \sum_{i=1}^M a_i z^{-i} Y(z)$$

That is:

$$\begin{aligned} Y(z) &= \sum_{i=0}^N b_i z^{-i} X(z) + \sum_{i=1}^M a_i z^{-i} Y(z) \\ Y(z) - \sum_{i=1}^M a_i z^{-i} Y(z) &= \sum_{i=0}^N b_i z^{-i} X(z) \end{aligned}$$

$$\frac{Y(z)}{X(z)} = \frac{\sum_{i=0}^N b_i z^{-i}}{1 - \sum_{i=1}^M a_i z^{-i}} \quad (11)$$

This is called a transfer function and it is crucial for analyzing the filter's behaviour. Recall the definition of [DFT](#):

$$z_k = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-kj} y_k = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \left(e^{\frac{2\pi i j}{N}} \right)^{-k} y_k$$

Now we denote $z = e^{\frac{2\pi i j}{N}}$ and drop the normalization factor in front. The expression becomes:

$$\sum_{k=0}^{N-1} z^{-k} y_k$$

is exactly the Z-transform of the signal y (embedded into the infinite sequence). In general as N gets large we may assume that for any $\omega \in [0, 2\pi]$ exists $j \in \mathbb{N}$ such that $\frac{2\pi j}{N} \approx \omega$, which allows us to compute the presence of any frequency in the range $[0, 2\pi]$ by evaluating the signal Z-transform at a point $z = e^{i\omega}$. Moreover the Theorem [4.2](#) suggests that the function $H(z)$ represents an impulse response as it essentially does deconvolution in the frequency (z) domain.

Lemma 5.2 (Partial fraction expansion). Let $P(x) \in \mathbb{C}_M[x]$, $a_i \in \mathbb{C}$ for $i \in \{1, \dots, N\}$ and $M < N$

$$Q(x) = \frac{P(x)}{\prod_{k=1}^N (1 - a_k x)}$$

with $a_i \neq a_j \forall i \neq j$ and $a_i \neq 0$.

Then the function $Q(x)$ can be expressed in the following form:

$$Q(x) = \sum_{i=1}^N \frac{A_i}{(1 - a_i x)}$$

and moreover this representation is unique.

Proof. Let $Q_i(x) = \prod_{k \neq i}^N (1 - a_k x)$. Consider their linear combination:

$$\sum_{i=1}^N \lambda_i Q_i(x) = 0$$

Substitute a_j^{-1} , which is well defined since $a_j \neq 0 \forall j \in \mathbb{N}$. We observe that $Q_i(a_j^{-1}) = 0$ for $i \neq j$ since Q_i contains the term $(1 - a_j x)$. Also $a_i a_j^{-1} \neq 1 \forall i \neq j$ since $a_i \neq a_j$, so $Q_j(a_j^{-1}) \neq 0$. Then:

$$\sum_{i=1}^N \lambda_i Q_i(a_j^{-1}) = \lambda_j Q_j(a_j^{-1}) = 0$$

Since the field $(\mathbb{C}, +, \cdot)$ is zero divisor free we must have $\lambda_j = 0$. This holds for all $j \in \{1, \dots, N\}$.

Thus Q_i 's form the basis of $\mathbb{C}_{N-1}[x]$ as $\dim \mathbb{C}_{N-1}[x] = N$, hence in particular we can decompose every polynomial in $\mathbb{C}_M[x]$ using the linear combination of Q_i :

$$\frac{P(x)}{\prod_{k=1}^N (1 - a_k x)} = \frac{\sum_{i=1}^N A_i Q_i}{\prod_{k=1}^N (1 - a_k x)} = \sum_{i=1}^N \frac{A_i Q_i}{\prod_{k=1}^N (1 - a_k x)} = \sum_{i=1}^N \frac{A_i}{1 - a_i x}$$

Thus we proved the existence of partial fraction decomposition whenever the numerator degree is less than the degree of denominator and denominator does not contain zero roots. Uniqueness follows from the basis properties. $\square$

Lemma 5.3 (Long division). Let $P \in C_N[x]$ and $Q \in C_M[x]$ with $N \geq M > 0$ and $\deg P = N$, $\deg Q = M$. Then there exists a polynomial $q \in C_{N-M}[x]$ and $r \in C_{M-1}[x]$, s.t. $P = q \cdot Q + r$

Proof. Let $P(x) = a_N x^N + \dots + a_1 x + a_0$ and $Q(x) = b_M x^M + \dots + b_1 x + b_0$ with $a_N \neq 0$ and $b_M \neq 0$ to ensure that polynomials have the proper degree. We prove this statement by induction:

Base: Let $N = M$. Set $q(x) = a_N/b_M$ and $r(x) = P(x) - q(x) \cdot Q(x)$. This fraction is well defined since $b_M \neq 0$. $P = q \cdot Q + r$ follows by construction, so we need to show that $\deg r(x) \leq M$:

$$\begin{aligned} r(x) &= a_N x^N + \dots + a_1 x + a_0 - \frac{a_N}{b_M} (b_M x^M + \dots + b_1 x + b_0) = \\ &= a_N x^N + \dots + a_1 x + a_0 - a_N x^M - \frac{a_N b_{M-1}}{b_M} x^{M-1} - \dots - \frac{a_N b_0}{b_M} \stackrel{N=M}{=} \\ &= a_N x^N + \dots + a_1 x + a_0 - a_N x^N - \frac{a_N b_{N-1}}{b_N} x^{N-1} - \dots - \frac{a_N b_0}{b_N} = \\ &= a_{M-1} x^{M-1} + \dots + a_1 x + a_0 - \frac{a_M b_{M-1}}{b_N} x^{M-1} - \dots - \frac{a_M b_0}{b_M} \end{aligned}$$

That is $\deg r \leq M - 1$

Step: Suppose the statement holds for some $N > M$. Then:

$$\begin{aligned} P(x) &= a_{N+1} x^{N+1} + \dots + a_1 x + a_0 = x(a_{N+1} x^N + \dots + a_2 x + a_1) + a_0 \\ &\stackrel{(a)}{=} x(q(x) \cdot Q(x) + r(x)) + a_0 = (xq(x)) \cdot Q(x) + (xr(x) + a_0) \end{aligned}$$

where (a) follows from the induction hypothesis, therefore $\deg q \leq N - M$ and $\deg r \leq M - 1$.

Note that $\deg xq \leq N + 1 - M$ and $\deg xr \leq M$, $\deg a_0 = 0$. If the degree of $xr(x) + a_0$ is exactly M we may use the base case to find the decomposition $xr(x) = \tilde{q}(x)Q(x) + \tilde{r}(x)$ where $\deg \tilde{q} = 0$ and $\deg \tilde{r} \leq M - 1$. Then $q^*(x) = xq(x) + \tilde{q}(x)$, $\deg q^* \leq N + 1 - M$ and $P(x) = q^*(x)Q(x) + \tilde{r}(x)$

$\square$

Theorem 5.3. Let $X(z)$ be a z -transform of a causal system $x[n]$ (i.e. $x[n] = 0, \forall n < 0$). Moreover assume that X is the rational function with distinct non-zero poles. Then $x[n]$ is absolutely summable if the pole with the maximum magnitude lies inside the unit circle.

Proof.

$$X(z) = \frac{P(z^{-1})}{Q(z^{-1})}$$

If $\deg P = N \geq M = \deg Q$, then the Lemma 5.3 ensures that exist $q(z^{-1}), r(z^{-1})$ with $\deg q \leq N - M$ and $\deg r \leq M - 1$, s.t. $P(z^{-1}) = q(z^{-1})Q(z^{-1}) + r(z^{-1})$. If $N < M$, then $q = 0$ and $r = P$

$$X(z) = \frac{q(z^{-1})Q(z^{-1}) + r(z^{-1})}{Q(z^{-1})} = q(z^{-1}) + \frac{r(z^{-1})}{Q(z^{-1})}$$

Let $d_1, \dots, d_M$ be the poles of X (distinct and non-zero by assumption), so $Q(z^{-1}) = \prod_{i=1}^M (1 - d_i z^{-1})$

Using the Lemma 5.2 we have that exist $A_1, \dots, A_M$, such that:

$$\begin{aligned} \frac{r(z^{-1})}{Q(z^{-1})} &= \sum_{i=1}^M \frac{A_i}{1 - d_i z^{-1}} \\ X(z) &= q(z^{-1}) + \sum_{i=1}^M \frac{A_i}{1 - d_i z^{-1}} = \sum_{i=0}^{N-M} c_i z^{-i} + \sum_{i=1}^M \frac{A_i}{1 - d_i z^{-1}} \end{aligned}$$

The assumption about causality allows us to uniquely determine the inverse z-transform by inspecting [the table](#).

$$\begin{aligned} c_i z^{-i} &\xleftrightarrow{\mathcal{Z}^{-1}} c_i \delta[n - i] \\ \frac{A_i}{1 - d_i z^{-1}} &\xleftrightarrow{\mathcal{Z}^{-1}} A_i d_i^n u[n] \end{aligned}$$

Using the Lemma 5.1.2:

$$\begin{aligned} \sum_{i=0}^{N-M} c_i z^{-i} + \sum_{i=1}^M \frac{A_i}{1 - d_i z^{-1}} &\xleftrightarrow{\mathcal{Z}^{-1}} \sum_{i=0}^{N-M} c_i \delta[n - i] + \sum_{i=1}^M A_i d_i^n u[n] \\ \sum_{n=0}^{\infty} |x[n]| &= \sum_{n=-\infty}^{\infty} \left| \sum_{i=0}^{N-M} c_i \delta[n - i] + \sum_{i=1}^M A_i d_i^n u[n] \right| \end{aligned}$$

Note that $\delta[n - i] = 1$ if and only if $n = i$ and zero elsewhere.

$$\begin{aligned} \sum_{n=-\infty}^{\infty} |x[n]| &\leq \sum_{n=-\infty}^{\infty} \sum_{i=0}^{N-M} |c_i \delta[n - i]| + \sum_{n=-\infty}^{\infty} \sum_{i=1}^M |A_i d_i^n u[n]| \\ &= \sum_{i=0}^{N-M} |c_i| + \sum_{n=-\infty}^{\infty} \sum_{i=1}^M |A_i d_i^n u[n]| \end{aligned}$$

By Theorem 5.1 and the proper extention of the finite sum to the infinite one (with zeros wherever the sequence is not defined):

$$\sum_{i=0}^{N-M} |c_i| + \sum_{n=-\infty}^{\infty} \sum_{i=1}^M |A_i d_i^n u[n]| = \sum_{i=0}^{N-M} |c_i| + \sum_{i=1}^M \sum_{n=-\infty}^{\infty} |A_i d_i^n u[n]|$$

And one can note that it converges if and only if

$$\sum_{n=-\infty}^{\infty} |A_i d_i^n u[n]| = |A_i| \sum_{n=0}^{\infty} |d_i^n| < \infty \quad \forall i \in \{1, \dots, M\}$$

This is the sum of geometric series, hence converges if and only if $|d_i| < 1$. We need this to hold for all $i \in \{1, \dots, M\}$, so $\max_i |d_i| < 1$ $\square$

Theorem 5.4. Let $x \xleftrightarrow{\mathcal{Z}} X(z)$ and $y \xleftrightarrow{\mathcal{Z}} Y(z)$, with both x, y being absolutely summable. Then $x * y \xleftrightarrow{\mathcal{Z}} X(z) \cdot Y(z)$

Proof.

$$h[n] = x * y = \sum_{k=-\infty}^{\infty} x[k]y[n-k]$$

By Theorem 5.1:

$$\begin{aligned} \sum_{n=-\infty}^{\infty} |h[n]| &= \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} |x[k]y[n-k]| \\ &\leq \sum_{k=-\infty}^{\infty} \left(|x[k]| \sum_{n=-\infty}^{\infty} |y[n-k]| \right) \\ &= \sum_{n=-\infty}^{\infty} |y[n]| \cdot \sum_{k=-\infty}^{\infty} |x[k]| < \infty \end{aligned}$$

By applying Theorem 5.2 and Lemma 5.1.1 (time shift property):

$$\begin{aligned} \sum_{n=-\infty}^{\infty} h[n]z^{-n} &= \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x[k]y[n-k]z^{-n} \\ &= \sum_{k=-\infty}^{\infty} x[k] \sum_{n=-\infty}^{\infty} y[n-k]z^{-n} \\ &= Y(z) \sum_{k=-\infty}^{\infty} x[k]z^{-k} \\ &= X(z)Y(z) \end{aligned}$$

$\square$

We say that z-transform converges if the series

$$\sum_{n=-\infty}^{\infty} |x[n]z^{-n}| = \sum_{n=-\infty}^{\infty} |x[n]|r^{-n}$$

converges¹. It can be easily seen that if the z transform converges for some point located at the distance r to the origin, then it converges for any point at the circle with radius r centered at the origin. This immediately yields the following equivalence:

$$X(e^{i\omega}) \text{ converges } \forall \omega \in [0, 2\pi] \Leftrightarrow x[n] \text{ absolutely summable}$$

¹eq. (3.7) [1]

Return to the Example 5.1:

$$y[n] = x_n + \alpha y[n - 1]$$

Its transfer function is:

$$H(z) = \frac{1}{1 - \alpha z^{-1}}$$

which has the only pole at $z = \alpha$, for which we require $|\alpha| < 1$, so that the ROC of H includes the unit circle.

By definition of the transfer function:

$$Y(z) = X(z)H(z)$$

Let $X \xleftrightarrow{Z^{-1}} y$ and $H \xleftrightarrow{Z^{-1}} h$, where both x and h are right-sided sequences.

Consider $(x * h)[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = 0$. For $n < 0$ we have that $n-k \geq 0 \Leftrightarrow k \leq n < 0$. In this case we have that $x[k] = 0$ since $k < 0$, so $(x * h)[n] = 0$.

By Theorem 5.4 we have that $X(z)H(z) \xleftrightarrow{Z^{-1}} x * h := y$ as we require y to be right-sided, so the inverse z transform is unique. If all the poles of the transfer function lie inside of the unit circle, then by Theorem 5.3 we have that h is absolutely summable. Moreover the Theorem 5.4 shows that the convolution of two absolutely summable sequences is absolutely summable as well. Therefore y is absolutely summable and hence the filter is stable.

Remark 5.1 (Sliding window average). Firstly we define the moving average of N points:

$$y[n] = \frac{1}{N} \sum_{k=n-N+1}^n x[k]$$

With the substitution $m = n - k$:

$$\frac{1}{N} \sum_{k=n-N+1}^n x[k] = \frac{1}{N} \sum_{m=0}^{N-1} x[n-m] = \sum_{m=0}^{N-1} h[m]x[n-m] = (x * h)[n]$$

so we define a kernel:

$$h[n] = \begin{cases} \frac{1}{N}, & 0 \leq n < N, \\ 0, & \text{otherwise} \end{cases}$$

We compute the transfer function:

$$h[n] \xleftrightarrow{Z} \sum_{k=-\infty}^{\infty} h[k]z^{-k} = \frac{1}{N} \sum_{k=0}^{N-1} z^{-k} = \frac{1}{N} \frac{1 - z^{-N}}{1 - z^{-1}}$$

which as (11) suggests corresponds to the following direct form:

$$y[n] = y[n-1] + \frac{1}{N}(x[n] - x[n-N])$$

Let us now plug in the definition of y :

$$\begin{aligned} y[n-1] + \frac{1}{N}(x[n] - x[n-N]) &= \frac{1}{N} \left(\sum_{k=n-N}^{n-1} x[k] + x[n] - x[n-N] \right) \\ &= \frac{1}{N} \sum_{k=n-N+1}^n x[k] = y[n] \end{aligned}$$

This is exactly the idea behind prefix sums (in particular using prefix sums to compute sliding window average).

5.2 Notch filter

We want to construct the transfer function with the zeros corresponding to the frequency ω . Its numerator then is the function $(z^{-1} - e^{i\omega})(z^{-1} - e^{-i\omega})$ since we need to cancel both positive and negative frequency oscillations. If we define the transfer function as is the frequency response looks as in the Figure fig:notch unscaled freq response

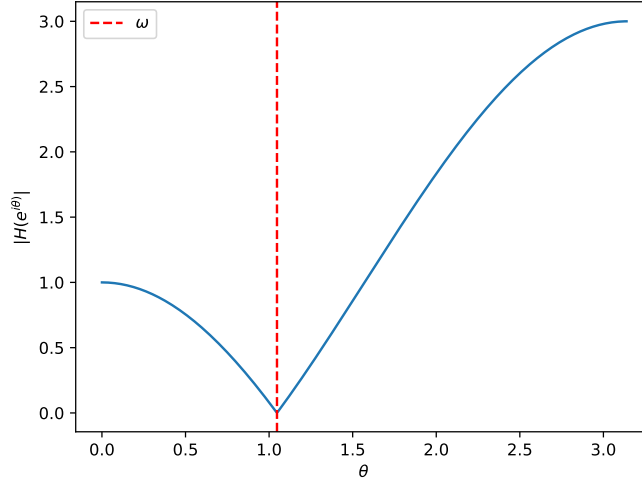


Figure 3: The frequency response of $(z^{-1} - e^{i\omega})(z^{-1} - e^{-i\omega})$

As we can see from the Figure 3 this filter produces a significant unevenness in the spectrum. Moreover we cannot really control the steepness of the constructed filter. To mitigate this we divide the transfer function we got previously by $(rz^{-1} - e^{i\omega})(rz^{-1} - e^{-i\omega})$. This helps to flatten the filter since $rz^{-1} - e^{i\omega} \approx (z^{-1} - e^{i\omega})$ for r reasonably close to 1. Note that dividing the transfer function by this expression adds two poles at $d_1 = re^{-i\omega}$ and $d_2 = re^{i\omega}$, so for the filter to be stable (converge for the unit circle) the radius r must be less than 1 by the Theorem 5.3. The bigger the value of r , the closer the pole is to the zeros, so it makes the filter steeper as it "lifts" the surface around it.

Define:

$$\begin{aligned} H(z) &= \frac{(z^{-1} - e^{i\omega})(z^{-1} - e^{-i\omega})}{(rz^{-1} - e^{i\omega})(rz^{-1} - e^{-i\omega})} = \frac{z^{-2} - (e^{-i\omega} + e^{i\omega})z^{-1} + 1}{r^2 z^{-2} - r(e^{-i\omega} + e^{i\omega})z^{-1} + 1} \\ &= \frac{z^{-2} - 2\cos(\omega)z^{-1} + 1}{r^2 z^{-2} - 2r\cos(\omega)z^{-1} + 1} \end{aligned}$$

Note that there are two spike slightly inside of the red circle and two dips. Those are the poles and the zeros of H .

And its frequency response looks as in the Figure 5

Note that for the transfer function $H(z)$ we have:

$$\overline{H(z)} = \frac{\overline{z^{-2} - 2\cos(\omega)z^{-1} + 1}}{\overline{r^2 z^{-2} - 2r\cos(\omega)z^{-1} + 1}} = \frac{\overline{z^{-2}} - 2\cos(\omega)\overline{z^{-1}} + 1}{r^2 \overline{z^{-2}} - 2r\cos(\omega)\overline{z^{-1}} + 1} = H(\bar{z})$$

That means for the frequency $e^{-i\theta}$ the frequency response has the same magnitude as for $e^{i\theta}$, but inverted phase. By the Remark 2.1.2, we have that the phase of the frequency response directly

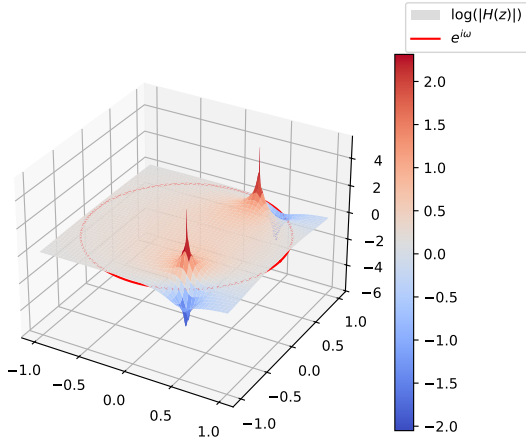


Figure 4: Notch filter topology

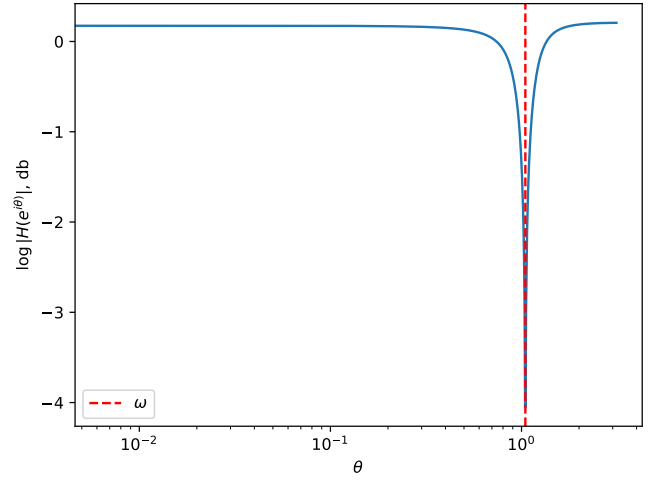


Figure 5: Notch filter frequency response

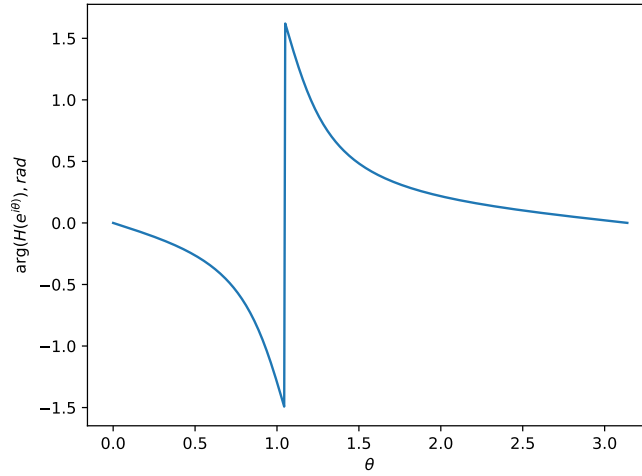


Figure 6: The notch filter phase response

relates to the corresponding oscillation phase. This allows us to visualize also the phase response of the constructed filter (Figure 6)

In a similar fashion we can build a peak filter by moving the zero of the filter closer to the origin, so that the pole influences the unit circle (frequency response) than the zeros do.

$$H(z) = \frac{r_1^2 z^{-2} - 2r_1 \cos(\omega) z^{-1} + 1}{r_2^2 z^{-2} - 2r_2 \cos(\omega) z^{-1} + 1} \quad r_1 < r_2 < 1$$

The notch filter then is the special case, where $r_1 = 1$.

6 Minimum phase

6.1 Group delay

This subsection solves the Exercise 5.57 [1]. Consider the signal:

$$x[n] = s[n] \cos(\omega_0 n)$$

which has the following z-transform at $e^{i\omega}$:

$$\begin{aligned} X(e^{i\omega}) &= \sum_{n=-\infty}^{\infty} s[n] \cos(\omega_0 n) e^{-i\omega n} = \sum_{n=-\infty}^{\infty} s[n] \cdot \frac{e^{i\omega_0 n} + e^{-i\omega_0 n}}{2} \cdot e^{-i\omega n} \\ &= \sum_{n=-\infty}^{\infty} s[n] \cdot \frac{e^{i(-\omega+\omega_0)n} + e^{i(-\omega-\omega_0)n}}{2} \\ &\stackrel{*}{=} \sum_{n=-\infty}^{\infty} \frac{s[n]}{2} e^{-i(\omega-\omega_0)n} + \sum_{n=-\infty}^{\infty} \frac{s[n]}{2} e^{-i(\omega+\omega_0)n} \\ &= \frac{1}{2} (S(e^{i(\omega-\omega_0)}) + S(e^{i(\omega+\omega_0)})) \end{aligned} \tag{12}$$

where we assume that the Fourier transform of $s[n]$ exists, so that in the step $*$ we are sure that both sums exist.

Assumption 1 $s[n]$ and $y[n]$ are real.

Assumption 2 $S(e^{-i\omega}) = 0$, $|\omega| > \Delta$ for some small $\Delta : \Delta \ll \omega_0, \Delta \ll \pi - \omega_0$ ²

1. Consider the phase response as in Figure 7 and $|H(e^{i\omega})| = 1$.

“ $\omega < 0$ ”:

We have that $\omega - \omega_0 < -\omega_0 < -\Delta$, so by the [Assumption 2](#) we have that:

$$X(e^{i\omega}) = \frac{1}{2} S(e^{i(\omega+\omega_0)})$$

Thus:

$$Y(\omega) = H(\omega)X(\omega) = \frac{1}{2} e^{i\phi_0} S(e^{i(\omega+\omega_0)})$$

“ $\omega \geq 0$ ”:

We have that $\omega + \omega_0 > \omega_0 > \Delta$, so by the [Assumption 2](#) we have that:

$$X(e^{i\omega}) = \frac{1}{2} S(e^{i(\omega-\omega_0)})$$

Thus:

$$Y(\omega) = H(\omega)X(\omega) = \frac{1}{2} e^{-i\phi_0} S(e^{i(\omega-\omega_0)})$$

²Those two inequalities ensure that the signal is spread enough so that it does not overlap with itself

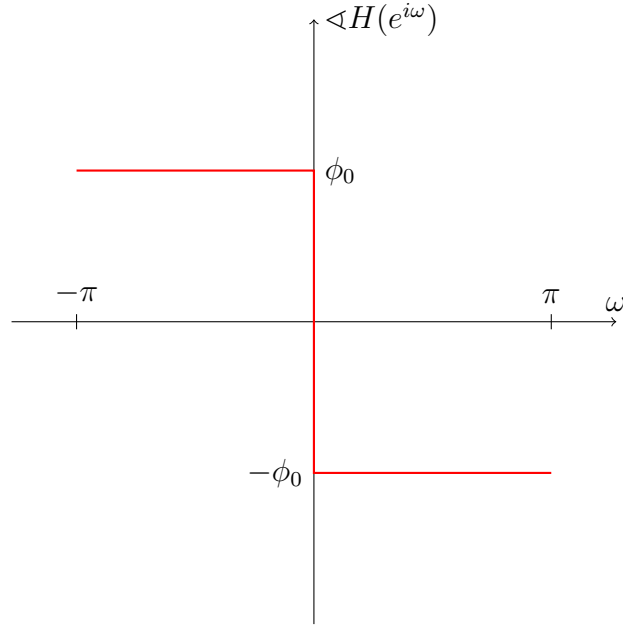


Figure 7

We take the inverse Fourier transform:

$$\begin{aligned}
2\pi \cdot y[k] &= \int_{-\pi}^{\pi} Y(\omega) e^{i\omega k} d\omega \\
&= \frac{1}{2} \left(\int_{-\pi}^0 e^{i\phi_0} S(e^{i(\omega+\omega_0)}) e^{i\omega k} d\omega + \int_0^{\pi} e^{-i\phi_0} S(e^{i(\omega-\omega_0)}) e^{i\omega k} d\omega \right) \\
&\stackrel{*}{=} \frac{1}{2} \left(\int_{-\pi+\omega_0}^{\omega_0} e^{i\phi_0} S(e^{i\omega}) e^{i\omega k} e^{-i\omega_0 k} d\omega + \int_{-\omega_0}^{\pi-\omega_0} e^{-i\phi_0} S(e^{i\omega}) e^{i\omega k} e^{i\omega_0 k} d\omega \right) \\
&= \frac{1}{2} \left(e^{-i(\omega_0 k - \phi_0)} \int_{-\pi+\omega_0}^{\omega_0} S(e^{i\omega}) e^{i\omega k} d\omega + e^{i(\omega_0 k - \phi_0)} \int_{-\omega_0}^{\pi-\omega_0} S(e^{i\omega}) e^{i\omega k} d\omega \right)
\end{aligned}$$

assuming Δ is small enough so that $\Delta \leq \pi - \omega_0$ we obtain that both integrals include the entire interval where $S(e^{i\omega})$ is non-zero hence we can extend them to $[-\pi, \pi]$:

$$\begin{aligned}
&\frac{1}{2} \left(e^{-i(\omega_0 k - \phi_0)} \int_{-\pi+\omega_0}^{\omega_0} S(e^{i\omega}) e^{i\omega k} d\omega + e^{i(\omega_0 k - \phi_0)} \int_{-\omega_0}^{\pi-\omega_0} S(e^{i\omega}) e^{i\omega k} d\omega \right) \\
&= \frac{1}{2} (e^{-i(\omega_0 k - \phi_0)} + e^{i(\omega_0 k - \phi_0)}) \int_{-\pi}^{\pi} S(e^{i\omega}) e^{i\omega k} d\omega = \frac{1}{2} \cdot 2 \cos(\omega_0 k - \phi_0) \cdot 2\pi \cdot s[k]
\end{aligned}$$

dividing both sides by 2π we get:

$$y[k] = s[k] \cos(\omega_0 k - \phi_0)$$

2. Consider the phase response as in Figure 7 and $|H(e^{i\omega})| = 1$.

Thus:

$$\angle H(\omega) = \begin{cases} \phi_0 - n_d \omega, & -\pi \leq \omega \leq 0 \\ -\phi_0 - n_d \omega, & 0 < \omega \leq \pi \end{cases}$$

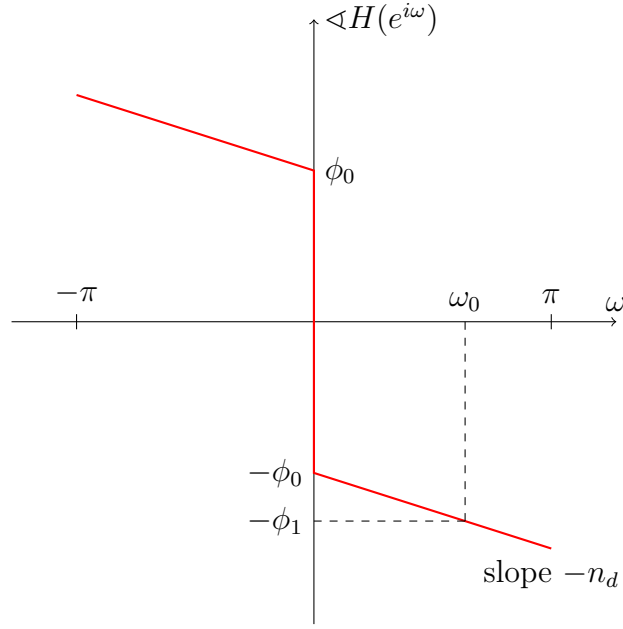


Figure 8

Using the result from part 1:

$$Y(\omega) = \begin{cases} \frac{1}{2}e^{i(\phi_0 - n_d\omega)}S(e^{i(\omega + \omega_0)}), & -\pi \leq \omega \leq 0 \\ \frac{1}{2}e^{i(-\phi_0 - n_d\omega)}S(e^{i(\omega - \omega_0)}), & 0 < \omega \leq \pi \end{cases}$$

By integrating the lower and upper halves of unit circle we get:

- Lower:

$$\begin{aligned} \int_{-\pi}^0 Y(\omega) e^{i\omega k} d\omega &= \frac{1}{2} \int_{-\pi}^0 e^{i(\phi_0 - n_d\omega)} S(e^{i(\omega + \omega_0)}) e^{i\omega k} d\omega \\ &\stackrel{*}{=} \frac{1}{2} \int_{-\pi + \omega_0}^{\omega_0} e^{i\phi_0} S(e^{i\omega}) e^{-in_d(\omega - \omega_0)} e^{i(\omega - \omega_0)k} d\omega \\ &= \frac{1}{2} \int_{-\pi + \omega_0}^{\omega_0} e^{i(\phi_0 + n_d\omega_0 - \omega_0 k)} S(e^{i\omega}) e^{-in_d\omega} e^{i\omega k} d\omega \\ &\stackrel{**}{=} \frac{1}{2} e^{i(\phi_0 + n_d\omega_0 - \omega_0 k)} \int_{-\pi}^{\pi} S(e^{i\omega}) e^{-in_d\omega} e^{i\omega k} d\omega \\ &\stackrel{***}{=} \frac{1}{2} e^{i(\phi_0 + n_d\omega_0 - \omega_0 k)} \cdot 2\pi \cdot s[k - n_d] \end{aligned}$$

where in * we used the change of variables $\omega \rightarrow \omega - \omega_0$, in ** we used the assumption [Assumption 2](#) and in *** we used the time shift property:

$$S(e^{i\omega}) e^{-in_d\omega} = \sum_{n=-\infty}^{\infty} s[n] e^{-i\omega n} e^{-in_d\omega} = \sum_{n=-\infty}^{\infty} s[n] e^{-i\omega(n + n_d)} \stackrel{\dagger}{=} \sum_{n=-\infty}^{\infty} s[n - n_d] e^{-i\omega n}$$

where in $\dagger$ we did the change of variables $n \rightarrow n + n_d$.

- Doing the same procedure as above we get:

$$\int_0^\pi Y(\omega) e^{i\omega k} = \frac{1}{2} e^{-i(\phi_0 + n_d \omega_0 - \omega_0 k)} \cdot 2\pi \cdot s[k - n_d]$$

By summing those two equation and dividing by 2π we obtain:

$$y[k] = s[k - n_d] \cos(\omega_0 k - \phi_0 - n_d \omega_0)$$

and by using line equation to express $-\phi_1 = -\phi_0 - \omega_0 n_d$ we obtain the equivalent form:

$$y[n] = s[n - n_d] \cos(\omega_0 n - \phi_1) \quad (13)$$

3. Since $x[n]$ is narrowband we have that $X(\omega)$ is non-zero in the neighbourhood $\omega_0 \pm \Delta$ and $-\omega_0 \pm \Delta$ so for the output signal we care only about those parts as $Y(\omega) = X(\omega)H(\omega)$. Using the basic property of z-transform $\bar{x} \xleftrightarrow{Z} \overline{X(\bar{z})}$ we get that for the real valued signal the phase is inverted when the argument is conjugated as $x = \bar{x} \xleftrightarrow{Z} \overline{X(\bar{z})} \Rightarrow X(\bar{z}) = \overline{X(\bar{z})} = \overline{X(z)}$. As we assume both signals x and y to be real valued this property automatically holds for H

In the neighbourhood of ω_0 the linear approximation takes the form:

$$\angle H(\omega) \approx \angle H(e^{i\omega_0}) + \partial_\omega(\angle H)(e^{i\omega_0}) \cdot (\omega - \omega_0)$$

Define $\tau_{gr} = -\partial_\omega(\angle H)(e^{i\omega_0})$ and $\tau_{ph} = -(1/\omega_0)\angle H(e^{i\omega_0})$. In terms of the Figure 8 it becomes $-\phi_1 = \angle H(e^{i\omega_0}) = -\omega_0 \tau_{ph}$ and slope $= -n_d = -\tau_{gr} \Rightarrow n_d = \tau_{gr}$. Plugging this into (13):

$$y[n] = s[n - n_d] \cos(\omega_0 n - \phi_1) = s[n - \tau_{gr}] \cos(\omega_0 n - \omega_0 \tau_{ph}) = s[n - \tau_{gr}] \cos(\omega_0(n - \tau_{ph})) \quad (14)$$

6.2 All-pass

Definition 6.1. We define a single pole all-pass system:

$$H_{ap}(z) = \frac{z^{-1} - \bar{a}}{1 - az^{-1}}$$

Lemma 6.1 (Properties of all-pass systems). Suppose $a = re^{i\theta}$, where $0 \leq r < 1$. Then:

1. $|H_{ap}(e^{i\omega})| = 1 \forall \omega \in [0, 2\pi]$
2. $\text{grd}(\omega) = \frac{1-r^2}{1-2r \cos(\omega-\theta)+r^2}$

Proof.

- 1.

$$\frac{(e^{i\omega})^{-1} - \bar{a}}{1 - a(e^{i\omega})^{-1}} = \frac{e^{-i\omega} - \bar{a}}{1 - ae^{-i\omega}} = e^{-i\omega} \frac{1 - \bar{a}e^{i\omega}}{1 - ae^{-i\omega}}$$

Note that:

$$\overline{1 - \bar{a}e^{i\omega}} = 1 - ae^{-i\omega}$$

so:

$$\left| \frac{1 - \bar{a}e^{i\omega}}{1 - ae^{-i\omega}} \right| = 1$$

thus:

$$\left| e^{-i\omega} \frac{1 - \bar{a}e^{i\omega}}{1 - ae^{-i\omega}} \right| = |e^{-i\omega}| \left| \frac{1 - \bar{a}e^{i\omega}}{1 - ae^{-i\omega}} \right| = 1$$

2. We compute:

$$\angle \left(\frac{(e^{i\omega})^{-1} - \bar{a}}{1 - a(e^{i\omega})^{-1}} \right) = \angle e^{-i\omega} + \angle \left(\frac{1 - \bar{a}e^{i\omega}}{1 - ae^{-i\omega}} \right) = \angle e^{-i\omega} + 2\angle(1 - \bar{a}e^{i\omega})$$

Then:

$$1 - \bar{a}e^{i\omega} = 1 - re^{-i\theta}e^{i\omega} = 1 - r\cos(\omega - \theta) - ir\sin(\omega - \theta)$$

Since $\text{Re}(1 - \bar{a}e^{i\omega}) > 0$, the point lies in the right half-plane, where $\angle z = \arctan(\text{Im } z / \text{Re } z)$:

$$\angle(1 - \bar{a}e^{i\omega}) = \arctan \left(\frac{-r\sin(\omega - \theta)}{1 - r\cos(\omega - \theta)} \right) = -\arctan \left(\frac{r\sin(\omega - \theta)}{1 - r\cos(\omega - \theta)} \right)$$

hence:

$$\angle \left(\frac{(e^{i\omega})^{-1} - \bar{a}}{1 - a(e^{i\omega})^{-1}} \right) = -\omega - 2\arctan \left(\frac{r\sin(\omega - \theta)}{1 - r\cos(\omega - \theta)} \right)$$

Now we compute the group delay:

Step 1.

$$\begin{aligned} \partial_\omega \left(\frac{r\sin(\omega - \theta)}{1 - r\cos(\omega - \theta)} \right) &= \frac{r\cos(\omega - \theta) \cdot (1 - r\cos(\omega - \theta)) - r\sin(\omega - \theta) \cdot r\sin(\omega - \theta)}{(1 - r\cos(\omega - \theta))^2} \\ &= \frac{r\cos(\omega - \theta) - r^2\cos^2(\omega - \theta) - r^2\sin^2(\omega - \theta)}{(1 - r\cos(\omega - \theta))^2} \\ &= \frac{r\cos(\omega - \theta) - r^2}{(1 - r\cos(\omega - \theta))^2} \end{aligned}$$

Step 2.

$$\begin{aligned} 1 + \left(\frac{r\sin(\omega - \theta)}{1 - r\cos(\omega - \theta)} \right)^2 &= 1 + \frac{r^2\sin^2(\omega - \theta)}{1 - 2r\cos(\omega - \theta) + r^2\cos^2(\omega - \theta)} \\ &= \frac{1 - 2r\cos(\omega - \theta) + r^2\cos^2(\omega - \theta) + r^2\sin^2(\omega - \theta)}{1 - 2r\cos(\omega - \theta) + r^2\cos^2(\omega - \theta)} \\ &= \frac{1 - 2r\cos(\omega - \theta) + r^2}{(1 - r\cos(\omega - \theta))^2} \end{aligned}$$

Step 3.

$$\begin{aligned} \partial_\omega \arctan \left(\frac{r\sin(\omega - \theta)}{1 - r\cos(\omega - \theta)} \right) &= \frac{1}{1 + \left(\frac{r\sin(\omega - \theta)}{1 - r\cos(\omega - \theta)} \right)^2} \cdot \partial_\omega \left(\frac{r\sin(\omega - \theta)}{1 - r\cos(\omega - \theta)} \right) \\ &= \frac{(1 - r\cos(\omega - \theta))^2}{1 - 2r\cos(\omega - \theta) + r^2} \cdot \frac{r\cos(\omega - \theta) - r^2}{(1 - r\cos(\omega - \theta))^2} \\ &= \frac{r\cos(\omega - \theta) - r^2}{1 - 2r\cos(\omega - \theta) + r^2} \end{aligned}$$

Step 4.

$$\begin{aligned}
\partial_\omega \left(-\omega - 2 \arctan \left(\frac{r \sin(\omega - \theta)}{1 - r \cos(\omega - \theta)} \right) \right) &= -1 - 2 \cdot \partial_\omega \arctan \left(\frac{r \sin(\omega - \theta)}{1 - r \cos(\omega - \theta)} \right) \\
&= -1 - \frac{2r \cos(\omega - \theta) - 2r^2}{1 - 2r \cos(\omega - \theta) + r^2} \\
&= \frac{r^2 - 1}{1 - 2r \cos(\omega - \theta) + r^2}
\end{aligned}$$

Thus:

$$\text{grd}(\omega) = -\partial_\omega(\angle H_{ap}(e^{i\omega})) = \frac{1 - r^2}{1 - 2r \cos(\omega - \theta) + r^2}$$

which is non-negative since $0 \leq r < 1$, hence $-2r \cos(\omega - \theta) \geq -2r$, so:

$$1 - 2r \cos(\omega - \theta) + r^2 \geq 1 - 2r + r^2 = (1 - r)^2 > 0$$

□

By cascading two all-pass filters with the conjugate poles we get:

$$H_{ap}(z) = \frac{z^{-1} - \bar{a}}{1 - az^{-1}} \frac{z^{-1} - a}{1 - \bar{a}z^{-1}} = \frac{z^{-2} - (\bar{a} + a)z^{-1} + |a|^2}{1 - (\bar{a} + a)z^{-1} + |a|^2 z^{-2}}$$

with all the real coefficients as $a + \bar{a} = 2 \text{Re}(a)$

By Lemma 2 we have that the group delay of the both parts is non-negative, hence the group delay of the whole system is non-negative as well as the phases add up. By Mean Value Theorem:

$$\frac{\angle H_{ap}(e^{i\omega}) - \angle H_{ap}(e^{i\cdot 0})}{\omega} = \partial_\omega \angle H_{ap}(e^{i\omega'}) \leq 0$$

for some $\omega' \in (0, \omega)$

$$H_{ap}(1) = \frac{1 - 2 \text{Re}(a) + |a|^2}{1 - 2 \text{Re } a + |a|^2} = 1 \Rightarrow \angle H_{ap}(e^{i\cdot 0}) = 0$$

thus:

$$\angle H_{ap}(e^{i\omega}) = \omega \cdot \partial_\omega \angle H_{ap}(e^{i\omega'}) \leq 0 \quad \forall \omega \in [0, \pi]$$

6.3 All-pass decomposition

Let H be a stable causal system with real-valued coefficients. Then for any zero it holds that its conjugate must be zero as well. Suppose H has a zero outside of the unit circle, namely at $z = 1/(c)$. Then it has a zero at $z = 1/(\bar{c})$ which we can factor out:

$$\begin{aligned}
H(z) &= \tilde{H}(z) \cdot (1 - 1/c \cdot z^{-1})(1 - 1/\bar{c} \cdot z^{-1}) \\
&= \frac{1}{|c|^2} \tilde{H}(z) \cdot (c - z^{-1})(\bar{c} - z^{-1}) \\
&= \frac{1}{|c|^2} \tilde{H}(z) \cdot \frac{1 - cz^{-1}}{1 - cz^{-1}} (c - z^{-1}) \cdot \frac{1 - \bar{c}z^{-1}}{1 - \bar{c}z^{-1}} (\bar{c} - z^{-1}) \\
&= \frac{1}{|c|^2} \tilde{H}(z) \cdot (1 - cz^{-1})(1 - \bar{c}z^{-1}) \cdot \frac{c - z^{-1}}{1 - \bar{c}z^{-1}} \cdot \frac{\bar{c} - z^{-1}}{1 - cz^{-1}}
\end{aligned}$$

Remark 6.1. The decomposition above holds for real-valued zeros too, but requires factoring out a single pole all-pass filter.

We have split the original system into the cascade of the system where the zeros $1/c$ and $1/\bar{c}$ become c and $\bar{c}$ and an all-pass system. By doing this for all the zeros outside of the unit-circle we get:

$$H = H_{min}(z) \cdot H_{ap}(z)$$

where $H_{ap}(z)$ is the cascade of the single all-pass filters. Using the result from Section 6.2 the phase of H_{ap} is non-positive, so $\angle H_{min}(e^{i\omega}) \geq \angle H(e^{i\omega}) \forall \omega \in [0, \pi]$, meaning that the phase delay $\tau_{ph} = -(1/\omega_0) \angle H_{min}(e^{i\omega_0}) \leq -(1/\omega_0) \angle H(e^{i\omega_0})$. With the same argumentation we can prove that the group delay decreases as well. When we fix the frequency response magnitude of the system we can reconstruct the original system up to its zeros. Since we can choose only from the original zeros of the system and the conjugate reciprocals (Section 5.4 [1]), we obtain the min-phase system by choosing only the zeros lying inside the unit circle. Otherwise with the argumentation above we can reduce phase and group delay by factoring out an all-pass system.

References

- [1] Alan V. Oppenheim and Ronald W. Schaffer. *Discrete-Time Signal Processing*. Pearson, Upper Saddle River, NJ, 3rd edition, 2010.